Макаж №2

1. $\int \frac{2x+11}{x^2+6x+13}dx = \int \frac{d(x^2+6x+13)}{x^2+6x+13} + 5\int \frac{d(x+3)}{(x+3)^2+4} = \ln|x^2+6x+13| + \frac{5}{2}\operatorname{arctg}\frac{x+3}{2} + C$

2. $\int \frac{4x^2+4x-11}{(2x-1)(2x+3)(2x-5)}dx = 2\int \frac{dx}{2x-1} - \frac{1}{4}\int \frac{dx}{2x+3} + \frac{3}{4}\int \frac{dx}{2x-5} = \ln|2x-1| - \frac{\ln|2x+3|}{8} \oplus$

$\sim$ $\frac{4x^2+4x-11}{(2x-1)(2x+3)(2x-5)} = \frac{A}{2x-1} + \frac{B}{2x+3} + \frac{C}{2x-5} = \frac{A(2x+3)(2x-5) + B(2x-1)(2x-5) + C(2x-1)(2x+3)}{(2x-1)(2x+3)(2x-5)}$

$x=2,5: 4\cdot 8\cdot C = 25+10-11 \Rightarrow 32C = 24 \Rightarrow C = 0,75 = \frac{3}{4}$

$x=0,5: -16A = -8 \Rightarrow A = 2$

$x=-1,5: +32B = -8 \Rightarrow B = -\frac{1}{4}$

$\sim$

$\oplus \frac{3}{8}\ln|2x-5| + C$

3. $\int \frac{5x-3}{(x-2)(3x^2+2x-1)}dx = \int \frac{5x-3}{(x-2)(3x-1)(x+1)}dx \ominus$

$\sim$ $3x^2+2x-1 = 3x^2+3x-x-1 = 3x(x+1) - (x+1) = (3x-1)(x+1)$

$\frac{5x-3}{(x+1)(3x-1)(x-2)} = \frac{A}{x+1} + \frac{B}{3x-1} + \frac{C}{x-2} = \frac{A(3x-1)(x-2) + B(x+1)(x-2) + C(x+1)(3x-1)}{(x+1)(3x-1)(x-2)}$

$x=2: 15C = 7 \Rightarrow C = \frac{7}{15}$

$x=-1: 12A = -8 \Rightarrow A = -\frac{2}{3}$

$x=\frac{1}{3}: -\frac{4}{3}\cdot\frac{5}{3}B = \frac{5}{3} - 3 \Rightarrow -20B = -4 \Rightarrow B = 5$

$\sim$ $\ominus -\frac{2}{3}\int \frac{dx}{x+1} + 5\int \frac{dx}{3x-1} + \frac{7}{15}\int \frac{dx}{x-2} = -\frac{2}{3}\ln|x+1| + \frac{5}{3}\ln|3x-1| + \frac{7}{15}\ln|x-2| + C$

4. $\int \frac{x^2+1}{x(x-1)^3}dx = -\int \frac{dx}{x} + \int \frac{dx}{x-1} + 2\int \frac{d(x-1)}{(x-1)^3} = -\ln|x| + \ln|x-1| - \frac{1}{(x-1)^2} + C$

$\sim$ $\frac{x^2+1}{x(x-1)^3} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{D}{(x-1)^3} = \frac{A(x-1)^3 + Bx(x-1)^2 + Cx(x-1) + Dx}{x(x-1)^3}$

$x=1: D=2$

$x=0: A=-1$

$\begin{cases} 2x = Cx + D = Cx + 2 \Rightarrow C = 0 \\ 2 = 2Bx \Rightarrow B = 1 \end{cases}$

$\sim$ 5. $\int \cos x\cos 4x\,dx = \frac{1}{2}\int \cos 5x\,dx + \frac{1}{2}\int \cos 3x\,dx = \frac{\sin 5x}{10} + \frac{\sin 3x}{6} + C$

$\cdot \cos x\cos 4x = \frac{1}{2}(\cos 5x + \cos 3x)$

6. $\int \frac{\cos^2 x}{\sin 4x}dx = \frac{1}{2}\int \frac{(1+\cos 2x)}{2\sin 2x\cos 2x}dx = \frac{1}{2}\int \frac{dx}{\sin 4x} + \frac{1}{4}\int \frac{dx}{\sin 2x} = \frac{1}{8}\int \frac{d4x}{\sin 4x} + \frac{1}{8}\int \frac{d2x}{\sin 2x} \ominus$

$\sim$ $\cdot \cos^2 x = \frac{1+\cos 2x}{2}$; $\int \frac{dx}{\sin x} = \int \frac{\sin x\,dx}{\sin^2 x} = -\int \frac{d\cos x}{1-\cos^2 x} = \int \frac{d\cos x}{\cos^2 x - 1} = \frac{1}{2}\ln\left|\frac{\cos x - 1}{\cos x + 1}\right| + C$

$\sim$

$= \frac{1}{16}\ln\left|\frac{\cos 4x-1}{\cos 4x+1}\right| + \frac{1}{16}\ln\left|\frac{\cos 2x-1}{\cos 2x+1}\right| + C$

7. $\int \cos^3 x \cos 8x\,dx =$

$\cos^3 x = \frac{\cos x}{2}(1+\cos 2x) = \frac{\cos x}{2} + \frac{\cos 2x \cos x}{2}$

$\cos x \cos 8x = \frac{1}{2}(\cos 9x + \cos 7x) = \frac{\cos 9x}{2} + \frac{\cos 7x}{2}$

$\cos 2x \cos x = \frac{1}{2}(\cos 3x + \cos x) = \frac{\cos 3x}{2} + \frac{\cos x}{2}$

$\cos 3x \cos 8x = \frac{1}{2}(\cos 11x + \cos 5x) = \frac{\cos 11x}{2} + \frac{\cos 5x}{2}$

$= \frac{1}{4}\int \cos 9x\,dx + \frac{1}{4}\int \cos 7x\,dx + \frac{1}{8}\int \cos 11x\,dx + \frac{1}{8}\int \cos 5x\,dx + \frac{1}{8}\int \cos 9x\,dx + \frac{1}{8}\int \cos 7x\,dx =$

$= \frac{3}{72} = \frac{1}{24}\sin 9x + \frac{3}{56}\sin 7x + \frac{1}{88}\sin 11x + \frac{1}{40}\sin 5x + C$

8. $\int \sin^4 x\,dx = \frac{1}{4}\int (1-2\cos 2x + \cos^2 2x)\,dx = \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{8}\int (1+\cos 4x)\,dx =$

$\sin^2 x = \frac{1-\cos 2x}{2}$; $\cos^2 2x = \frac{1+\cos 4x}{2}$

$= \frac{3}{8}x - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$ Ну гениально

10. $\int \frac{dx}{\sin 4x} = \frac{1}{4}\int \frac{d4x}{\sin 4x} = -\frac{1}{4}\int \frac{d\cos 4x}{1-\cos^2 4x} = \frac{1}{8}\ln\left|\frac{\cos 4x-1}{\cos 4x+1}\right| + C$

11. $\int \frac{dx}{\cos 2x - \sin 2x} = \int \frac{dx}{\cos^2 x \cdot (1-\operatorname{tg} x)^2 - 2\operatorname{tg}^2 x} = \int \frac{d\operatorname{tg} x}{1-2\operatorname{tg} x - \operatorname{tg}^2 x} = -\int \frac{d\operatorname{tg} x}{\operatorname{tg}^2 x + 2\operatorname{tg} x + 1 - 2} =$

$\cos 2x - \sin 2x = (\cos x - \sin x)^2 - 2\sin^2 x$

$= -\int \frac{d(\operatorname{tg} x + 1)}{(\operatorname{tg} x + 1)^2 - 2} = -\frac{1}{2\sqrt{2}}\ln\left|\frac{\operatorname{tg} x + 1 - \sqrt{2}}{\operatorname{tg} x + 1 + \sqrt{2}}\right| + C$

13. $\int e^{\arccos x}\,dx = \left|\begin{matrix} x = \sin t \\ dx = \cos t\,dt \end{matrix}\right| = \int e^t \cos t\,dt = \frac{e^t \cos t + e^t \sin t}{2} =$

$\int e^t \cos t\,dt = \left|\begin{matrix} u = \cos t,\ du = -\sin t\,dt \\ dv = e^t dt,\ v = e^t \end{matrix}\right| = \left(t = \arccos x\right) e^t \cos t + \int e^t \sin t\,dt = \left|\begin{matrix} u = \sin t,\ du = \cos t\,dt \\ dv = e^t dt,\ v = e^t \end{matrix}\right| =$

$= e^t \cos t + e^t \sin t - \int e^t \cos t\,dt \Rightarrow \int e^t \cos t\,dt = \frac{e^t \cos t + e^t \sin t}{2}$

$= \frac{e^{\arccos x}\cos \arccos x + e^{\arccos x} x}{2} = \frac{e^{\arccos x}\left(\frac{2}{\sqrt{1-x^2}} + x\right)}{2} + C$

$\cos \arccos x = \sqrt{1-x^2}$

9. $\int \frac{\cos^5 x}{\sin^3 x}\,dx = \int \frac{\cos^5 x \sin x}{\sin^4 x}\,dx = -\frac{1}{2}\int \frac{\cos^4 x\,d\cos^2 x}{(\cos^2 x - 1)^2} = \left|p = \cos^2 x\right| = -\frac{1}{2}\int \frac{p^2\,dp}{(p-1)^2} =$

$= -\frac{1}{2}\int \left(1 + \frac{1}{p-1}\right)^2 dp = -\frac{1}{2}\int \left(1 + \frac{2}{p-1} + \frac{1}{(p-1)^2}\right) dp = -\frac{p}{2} - \int \frac{dp}{p-1} - \frac{1}{2}\int \frac{dp}{(p-1)^2} = -\frac{p}{2} - \ln|p-1| + \frac{1}{2(p-1)} + C =$

$= -\frac{\cos^2 x}{2} - \ln|\cos^2 x - 1| + \frac{1}{2(\cos^2 x - 1)} + C = -\frac{\cos^2 x}{2} - \ln \sin^2 x - \frac{1}{2\sin^2 x} + C$

14. $\int \frac{dx}{x^6+1} = \int \frac{dx}{(x^2+1)(x^4-x^2+1)} = \int \frac{(x^2+1)-x^2}{(x^2+1)(x^4-x^2+1)}dx = \int \frac{dx}{x^4-x^2+1} - \int \frac{x^2 dx}{(x^2+1)(x^4-x^2+1)} =$

$\int \frac{x^2 dx}{(x^2+1)(x^4-x^2+1)} = \frac{1}{3}\int \frac{dx^3}{x^6+1} = \frac{1}{3}\int \frac{dx^3}{(x^3)^2+1} = \frac{1}{3}\operatorname{arctg} x^3 + C$

$\int \frac{dx}{x^4-x^2+1} = \int \frac{(x^2+1-x^2)dx}{x^4-x^2+1} = \int \frac{(x^2+1)dx}{x^4-x^2+1} - \int \frac{x^2 dx}{x^4-x^2+1} = \int \frac{(1+\frac{1}{x^2})dx}{x^2-1+\frac{1}{x^2}} - \int \frac{x^2 dx}{x^4-x^2+1} =$

$d(x-\frac{1}{x})$

$(x-\frac{1}{x})^2+1$

$= \operatorname{arctg}(x-\frac{1}{x}) - \int \frac{x^2 dx}{x^4-x^2+1} =$

$\int \frac{x^2 dx}{x^4-x^2+1} = \int \frac{x^2 dx}{(x^2-\sqrt{3}x+1)(x^2+\sqrt{3}x+1)} =$

$A = -C$

$B = -D$

$\frac{x^2}{(x^2-\sqrt{3}x+1)(x^2+\sqrt{3}x+1)} = \frac{Ax+B}{x^2-\sqrt{3}x+1} + \frac{Cx+D}{x^2+\sqrt{3}x+1} = \frac{(Ax+B)(x^2+\sqrt{3}x+1)+(Cx+D)(x^2-\sqrt{3}x+1)}{(x^2-\sqrt{3}x+1)(x^2+\sqrt{3}x+1)} \Rightarrow$

$\Rightarrow (Ax^3+Bx^2+\sqrt{3}Ax^2+\sqrt{3}Bx+Ax+B)+(Cx^3+Dx^2-\sqrt{3}Cx^2-\sqrt{3}Dx+Cx+D) = x^2$

$A = -C$

$B+\sqrt{3}A+D-\sqrt{3}C = 1 \Rightarrow -2\sqrt{3}C = 1 \Rightarrow C = -\frac{1}{2\sqrt{3}} = -\frac{\sqrt{3}}{6} \Rightarrow A = \frac{\sqrt{3}}{6}$

$\sqrt{3}B - \sqrt{3}D = 0 \Rightarrow B = D = 0$

$= \frac{\sqrt{3}}{6}\int \frac{x\,dx}{x^2-\sqrt{3}x+1} - \frac{\sqrt{3}}{6}\int \frac{x\,dx}{x^2+\sqrt{3}x+1} =$

$d(x^2-\sqrt{3}x+1)$

$\frac{\sqrt{3}}{6}\int \frac{x\,dx}{x^2-\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\int \frac{2x\,dx}{x^2-\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\int \frac{2x-\sqrt{3}}{x^2-\sqrt{3}x+1}dx + \frac{1}{4}\int \frac{dx}{x^2-\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\ln|x^2-\sqrt{3}x+1| +$

$+\frac{1}{4}\int \frac{d(x-\frac{\sqrt{3}}{2})}{(x-\frac{\sqrt{3}}{2})^2+\frac{1}{4}} = \frac{\sqrt{3}}{12}\ln|x^2-\sqrt{3}x+1| + \frac{\operatorname{arctg}(2x-\sqrt{3})}{2}$

$\frac{\sqrt{3}}{6}\int \frac{x\,dx}{x^2+\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\int \frac{2x\,dx}{x^2+\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\int \frac{2x-\sqrt{3}}{x^2+\sqrt{3}x+1}dx + \frac{1}{4}\int \frac{dx}{x^2+\sqrt{3}x+1} = \frac{\sqrt{3}}{12}\ln|x^2+\sqrt{3}x+1| +$

$+\frac{1}{4}\int \frac{dx}{(x+\frac{\sqrt{3}}{2})^2+\frac{1}{4}} = \frac{\sqrt{3}}{12}\ln|x^2+\sqrt{3}x+1| + \frac{\operatorname{arctg}(2x+\sqrt{3})}{2} + C$

$= \frac{\sqrt{3}}{12}\ln\left|\frac{x^2-\sqrt{3}x+1}{x^2+\sqrt{3}x+1}\right| + \frac{1}{2}(\operatorname{arctg}(2x-\sqrt{3}) - \operatorname{arctg}(2x+\sqrt{3})) + C$

$= \operatorname{arctg}\frac{x^2-1}{x} - \frac{1}{4\sqrt{3}}\ln\left|\frac{x^2-\sqrt{3}x+1}{x^2+\sqrt{3}x+1}\right| - \frac{\operatorname{arctg}(2x-\sqrt{3})}{2} + \frac{\operatorname{arctg}(2x+\sqrt{3})}{2} + C$

$= -\frac{\operatorname{arctg}(2x-\sqrt{3})}{2} + \frac{\operatorname{arctg}(2x+\sqrt{3})}{2} + \operatorname{arctg}\frac{x^2+1}{x} - \frac{1}{4\sqrt{3}}\ln\left|\frac{x^2-\sqrt{3}x+1}{x^2+\sqrt{3}x+1}\right| - \frac{\operatorname{arctg} x^3}{8} + C$